Supplementary Material: Map Growth and Policy Routing in Budgeted Online RGB-D 3DGS

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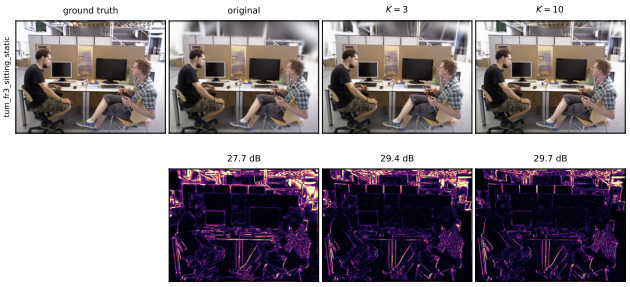


Figure S1. Second held-out view (`fr3_sitting_static`), as in Fig. S8 of the main paper.

A. Time-targeted steps: details

Per-scene results (Tab. S1). Mean PSNR and frame time for the original system, uniform $K{=}3$ and $K{=}10$, and the time-targeted controller with value-gated growth at 45 and 75 ms for each of the 11 test scenes and five fresh windows.

Closed-loop tracker. Each frame's pose is estimated by rendering the current map and optimizing a 6-DoF correction coarse to fine on a photometric plus depth L1 loss (render scales $1/4, 1/2, 1$ with $20/15/10$ Adam steps, learning rate 10^{-3}, constant-velocity initialization), without keyframes or loop closure. Tracking time (about 130–180 ms per frame) is not included in the mapping frame time. The local Replica sequences move 13–19 cm and 7–11° between frames (against 0.3–1.1 cm and 0.3–0.7° on TUM), so no tracker converges there.

Optimizer inside MonoGS's own mapper. Separately from the hybrid system of the main paper, we tried the two transferable ingredients, short-memory Adam and learning-rate scaling, and the mapping iteration count in MonoGS [**?**] (RGB-D, single thread, `fr1_desk`, first 150 frames, $640{\times}480$, no final color refinement, one run per setting; two identical default runs differed by about 0.15 dB and 0.1 cm). Default: 18.90 dB, ATE 1.57 cm, 136 s. Betas $(0.5, 0.8)$: 18.92 dB, 2.21 cm, 134 s. Betas plus learning rates $\times 3$: 19.25 dB, 1.39 cm, 142 s. Default with 50 instead of 150 mapping iterations: 18.72 dB, 1.44 cm, 140 s; betas with 50 iterations: 19.19 dB, 3.31 cm, 138 s. The changes are within noise for PSNR except at most $+0.35$ dB, trajectory error is worse in two settings, and total time barely moves because MonoGS spends it on tracking (100 pose iterations per frame) and a multi-keyframe mapping window rather than on a single-frame step loop. Changing the optimizer inside MonoGS's own multi-keyframe mapper therefore does not help; the benefit appears when the whole mapper is replaced, as in the hybrid system of the main paper (its MonoGS table). Raw numbers: `results/monogs_check/summary.json`.

Replay design on the development scenes. Final-map PSNR on never-mapped frames (ground-truth poses, four development scenes, one seed; Tab. **??** of the main paper uses the frozen test scenes). Original system 23.07 dB; $K{=}3$ without replay 23.77 dB; $K{=}3$ with the chosen replay (every third step, 32-frame reservoir, every second mapped frame offered) 25.35 dB. At $K{=}3$ the replayed step is the same for "every second" (25.39 dB) and "every third" steps, and "every fourth" never triggers (23.80 dB); reservoir sizes 16 and 64 give 25.03 and 25.37 dB, and offering every frame or every third gives 25.21 and 25.33 dB. At $K{=}10$, replaying every third step gives 25.44 dB (no replay: 22.61 dB), every second step 26.26 dB and every fifth 24.81 dB: a larger replay share retains more at the price of about 0.1 dB of next-frame PSNR, and we kept the constant fixed before the test evaluation. Prioritized replay (frames drawn in proportion to their last loss) gave 25.07 and 25.50 dB at $K{=}3$ and $K{=}10$, no better than uniform sampling. Drawing the two replayed frames of the joint step only from the 4 or 8 stored frames nearest in camera pose (a keyframe window by proximity, $K{=}3$, four development scenes, two seeds) is clearly worse than the uniform reservoir: 22.8 and 23.5 dB against 25.1 dB, and about 4 ms per frame slower; coverage of the whole history matters more than overlap with the current view. Forgetting is not caused by pruning: disabling pruning changes $K{=}3$ from 23.77 to 23.76 dB.

Joint replay design on the development scenes. Same measure and scenes as above, with measured frame time

Scene / window	orig.		K=3		K=10		T45		T75	
	dB	ms	dB	ms	dB	ms	dB	ms	dB	ms
test/replica_office0	30.59	47	37.99	34	41.96	70	40.09	46	42.02	74
test/replica_office1	33.87	45	39.41	34	43.73	71	41.28	46	43.76	74
test/replica_office3	24.35	47	33.51	36	37.89	77	35.21	46	37.80	75
test/replica_office4	25.45	47	35.05	35	39.65	71	37.36	46	39.62	75
test/replica_room0	22.58	48	29.89	35	34.61	72	32.03	46	34.63	75
test/replica_room2	25.57	46	33.33	36	38.27	75	35.19	46	38.18	75
test/tum_fr1_room	16.73	43	24.76	31	29.67	67	27.42	46	30.17	75
test/tum_fr1_rpy	21.11	39	29.70	27	36.33	57	34.85	45	36.60	74
test/tum_fr1_xyz	21.61	46	30.00	32	35.31	68	32.82	45	35.35	75
test/tum_fr2_xyz	24.23	44	32.94	30	37.54	64	35.90	45	37.79	74
test/tum_fr3_sitting_static	26.44	33	32.10	25	37.30	53	36.21	45	37.73	74
fresh/tum_fr1_desk	23.06	42	29.27	31	35.14	66	32.61	45	35.12	75
fresh/tum_fr1_room	16.83	45	22.96	30	29.56	61	27.23	45	29.33	73
fresh/tum_fr1_xyz	22.86	46	29.73	31	34.81	67	32.53	45	34.84	75
fresh/tum_fr2_xyz	24.62	39	31.86	28	36.65	58	35.42	45	37.25	74
fresh/tum_fr3_sitting_static	28.36	39	34.28	27	38.34	57	37.46	45	38.64	74

Table S1. **Per-scene results of the steps regime.** Mean PSNR (dB) and measured frame time (ms) per test scene and fresh window ("fresh/" rows), two seeds. T45 and T75 are time-targeted steps with value-gated growth at 45 and 75 ms targets.

(mean over the scenes). Sequential replay (every third step) plateaus as steps are added: 25.4 dB at K=3 (34 ms), 25.0 dB at K=5, 25.2 dB at K=6 and 25.0 dB at K=8 (62 ms). Joint steps (the first step on the current frame only, every later step on the current frame plus n stored frames) raise final-map PSNR at equal time: for instance K=2 with 3 stored frames (45 ms) 26.7 dB against 25.0 dB for sequential replay at K=5 (45 ms), K=3 with 2 stored frames (55 ms) 27.0 dB against 25.2 dB at K=6 (50 ms), and K=3 with 5 stored frames (84 ms) 27.8 dB. Next-frame PSNR is 0.1–0.9 dB lower than sequential replay at the same time. We froze K=3 with 2 stored frames before the test evaluation. Raw outputs: `results/retention_dev2`.

Development sweeps (Tab. S2). The optimizer constants came from coordinate search on the four development scenes at uniform K=8 steps with one seed. The Adam betas had by far the largest effect once the learning rates were raised; the learning-rate multiplier has a flat optimum near the chosen value. Per-group learning-rate multipliers of 0.5 or 2 changed the mean by at most 0.3 dB (inside seed noise) and were not adopted; within-frame decays other than $1.4 \to 0.6$ were no better (for example $1.8 \to 0.5$ and $1.2 \to 0.8$ lost 0.1 and 0.5 dB). Spawn rates of 250 and 1000 per frame and error-threshold multipliers of 0.5 and 2 were worse or flat, as were loss weights for SSIM (0.05, 0.4, 0.5, 1), depth (0.2, 1, 2) and color.

Exploratory negative results. These were run on the development scenes (one seed) before the betas were changed, so the optimizer is the learning-rate-only setting; the code was removed after the runs and the outputs are kept in `results/engine_dev` and `results/holdout_dev`. Differences are relative to the uniform-step curve at the method's own mean time. *Convergence-gated early stopping*: stop a frame's steps once one more step lowers the loss by less than a relative threshold δ (at least three steps, at most 16); δ=0.03, 0.015, 0.007 give -0.15, -0.29 and -0.30 dB. *Error-proportional allocation*: the step count of a frame is scaled by $(e_t/\bar{e})^{\beta}$ with e_t the pre-update color error and $\bar{e}$ its running mean (clipped to $[0.5, 2]$ or $[0.5, 3]$): β=0.5, 1, −0.5, −1 give -0.32, -0.39, -0.70, -1.04 dB against a local slope of 0.096 dB/ms. *Keyframe replay, recent frames, next-frame measure*: every third (or second) step optimizes a stored *recent* frame instead of the current one (this is not the reservoir replay of the main paper, which draws from the whole history and is evaluated on the final map); held-out PSNR changes by +0.23 (every third step, buffer 8, stride

Setting	**value**	PSNR	ms
Adam betas (β_1, β_2)	(0.9, 0.999) default	31.77	60
	(0.95, 0.999)	29.24	59
	(0.9, 0.99)	31.78	60
	(0.8, 0.99)	34.01	61
	(0.5, 0.99)	34.81	65
	(0.0, 0.99)	32.06	66
	(0.5, 0.9)	36.02	63
	(0.5, 0.8) chosen	36.07	63
	(0.5, 0.7)	36.03	63
	(0.3, 0.8)	35.56	64
	(0.5, 0.5)	35.74	63
learning-rate multiplier (betas 0.5, 0.8)	0.3	33.81	64
	0.5	35.26	64
	0.7	35.96	62
	1.0 chosen	36.07	63
	1.4	35.62	63
	2.0	34.17	62

Table S2. **Development-scene sweeps** behind the optimizer constants: mean PSNR (dB) and frame time over the four development scenes at uniform K=8 steps, one seed. The learning-rate sweep multiplies all rates of the preset.

2), +0.10 (every second step) and +0.30 dB (every third step, buffer 16), while mapped-frame PSNR falls by -3.36, -5.27 and -3.48 dB. `torch.compile` on the loss and the optimizer update did not change frame time.

Earlier controllers. "Prior TT+VGG" in Tab. **??** is the time-targeted *budget* (scaling the modeled budget by a multiplier $m \in [1, 16]$ to hold 100 ms) plus value-gated growth of our earlier version, evaluated on the same code; with the original optimizer it gains +2.59 dB, with the new optimizer +10.13 dB. In the steps regime, value-gated growth is neutral to slightly negative at equal time (0.03, -0.02, -0.17 and -0.11 dB at 45, 60, 75 and 90 ms on the test scenes). Its constants were not retuned. The closed-loop and count-cap analyses below used these earlier controllers on the previous code revision.

B. Earlier analysis of the original budgeted system

This material, from the earlier version of this work, was produced on the code revision before the core changes of Sec. **??** and with the original optimizer. It diagnoses why the original system's quality depends on the modeled budget ("compute dilution") and evaluates budget-level controllers; its qualitative conclusions about budget starvation are consistent with Sec. **??**, but its absolute numbers are not comparable with the main paper.

B.1. Setting and Method

Budgeted mapping. Let $G_t = \{g_1, \ldots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t. Per frame, a scheduler chooses a subset $A_t \subseteq G_t$ to optimize,

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \tag{1}$$

where ΔQ is the reconstruction improvement, approximated at runtime by a selector score, and $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ is a decoupled cost model for K microsteps. The budget B_t is *modeled* optimization time. **Measuring dilution.** We log two age-resolved statistics per frame: the *young budget share*, the fraction of modeled update budget spent on Gaussians younger than ten frames, and the *mature selection rate*, the fraction of Gaussians at least ten frames old that are selected. Dilution predicts a falling mature selection rate as the map grows. **Selection.** Candidates are scored by error × influence mass, optionally with temporal EMA filtering ($\beta = 0.90$), and packed by fractional-knapsack value density s_i/C_i under $B_t = 15$ ms with safety factor 1.10 (effective cap 13.6 ms). Each selected Gaussian receives $K = 5$ microsteps; the cost model is calibrated so modeled and wall-clock optimization time agree within measurement noise on the RTX 4050 backend. Exact Gaussian-to-pixel attribution costs 523.9 ms/frame, so we use a center-sampled projection, $\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4}\, e(\lfloor \mu_{2D,i} \rceil)$, that runs in 0.85 ms (614.5× faster) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. The scheduler totals 2.54 ms per frame. **Coverage throttling.** Coverage κ_t is the fraction of valid pixels with transmission < 0.5. The per-frame spawn cap ramps from 1.0 to 0.20 as κ_t goes from 0.80 to 0.90. A backlog limit exists but was inactive in the headline runs, because its count was only computed when warmup was enabled; the factorial evaluates it independently. Warmup pre-allocation is disabled and microsteps are fixed at $K = 5$.

B.2. Diagnosing Compute Dilution

Protocol. TUM RGB-D [**?**] and Replica [**?**] sequences at 320 × 240, $B_t = 15$ ms, seeds [42..49]; two-sided paired Wilcoxon tests, Holm correction and bootstrap CIs over per-seed trajectory summaries; production `gsplat` backend; ground-truth poses with tracking decoupled (a mapping benchmark, not SLAM ATE). With eight seeds the smallest attainable two-sided Wilcoxon p is 0.0078, and with fixed trajectories seeds measure repeatability only, so we report effect sizes with intervals and do not treat seed-level standardized effects as evidence of generalization. Sequences use the first 150 frames where available; the local Replica copies provide 99 evaluated frames for `office0` and 119 for `office1`. All results in Tabs. S3–S9 come from one code revision (git tag `paper-rerun-1`) with the corrected evaluator: standard Gaussian-window SSIM and alpha-normalized rendered depth. Earlier bundles used a global-statistics SSIM and a depth evaluator that biased depth L1 toward zero in partially transparent pixels, so their SSIM and depth values are not comparable with these. Artifacts: `results/rerun_paper1/`. **Holdout protocol.** Because online mapping optimizes on the evaluated trajectory, train-view PSNR overstates generalization. We hold out every 8th frame render-only (never mapped, never selected) and report held-out effects in Tab. S10. This is a same-trajectory proxy, not independent novel-trajectory generalization.

Main result (Tab. S3; Supp. Figs. S4–S5). OURS reaches 27.74 ± 0.03 dB: +2.36 over the strongest budgeted baseline, +4.21 over the RTG-SLAM-policy re-implementation, and +8.89 over pass-through. These results evaluate the combined policy on this trajectory.

Selector × throttle factorial (Tab. S4). With selector fixed to raw error, coverage adds +3.53 [3.38, 3.70] dB on `fr2_xyz` and +1.71 [1.64, 1.80] dB on `fr1_xyz`. With temporal×influence fixed, its gains are +2.49 [2.40, 2.59] and +1.50 [1.48, 1.52] dB. Brackets are paired seed-bootstrap 95% CIs; all four coverage contrasts have two-sided paired Wilcoxon $p = .0078$. Temporal selection itself adds +0.09 dB with coverage on `fr2_xyz` and +0.54 dB on

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.85	14.75	0.4655	67.2	21,128	+8.89 ✓
error_only	24.21	20.20	0.7023	11.0	72,366	+3.53 ✓
error_influence	25.38	22.35	0.7338	11.1	86,988	+2.36 ✓
error_influence_temporal	25.23	22.05	0.7315	10.9	86,187	+2.51 ✓
rtg_slam_reimpl	23.53	20.57	0.6833	10.1	60,050	+4.21 ✓
OURS	**27.74**	**28.73**	**0.7978**	**12.1**	**21,642**	—
FULL	*27.89*	*29.29*	*0.8036*	*11.3*	*70,295*	-0.15 ✓

Table S3. **Main benchmark, TUM `fr2_xyz`, 150 frames × 8 seeds.** All six controls differ significantly from OURS after Holm correction ($p_{\text{Holm}} = 0.0469$). FULL is 0.15 dB higher. Headroom recovery is 98.3% relative to no-op and 94.0% relative to the strongest budgeted baseline, at a 3.2× smaller map than FULL. Source: commit tag `paper-rerun-1`.

	TUM fr2_xyz			TUM fr1_xyz		
Selector / map control	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	24.13	.0205	73.1	21.52	.0378	79.1
Error, coverage	27.66	.0118	21.7	23.23	.0335	37.1
Error, coverage + backlog	27.67	.0117	21.5	23.32	.0334	35.1
Temporal×influence, no throttle	25.26	.0134	86.2	22.28	.0368	85.5
Temporal×influence, coverage	27.75	.0108	21.5	23.78	.0314	37.9
Temporal×influence, coverage + backlog	27.73	.0109	21.8	23.89	.0318	33.1
Error, optimization budget ×4	27.82	.0107	71.3	24.46	.0309	79.1
Error, hard map cap 23k	26.89	.0217	23.0	22.52	.1075	23.0
Error, hard map cap 39k	24.93	.0237	39.0	21.56	.1029	39.0

Table S4. **Corrected-pose factorial and dilution controls, 150 frames × 8 seeds per scene.** All factorial cells use the same modeled 15 ms optimization budget. The ×4 control uses 60 ms for optimization while keeping the original 15 ms spawn allowance. Depth L1 is in meters. Backlog is enabled independently of warmup only in the last throttle level. Per-run traces and paired intervals are in `results/rerun_paper1/dilution_factorial/`.

`fr1_xyz`; the interaction is negative (−1.04 and −0.22 dB), so selector and coverage effects are not additive. The independent backlog option adds +0.01/−0.02 dB on `fr2_xyz` and +0.08/+0.11 dB on `fr1_xyz` for error/temporal selection.

Dilution controls and direct allocation. Holding the spawn allowance at its 15 ms value, giving raw-error selection 60 ms for optimization raises PSNR by +3.69 [3.55, 3.83] and +2.94 [2.89, 3.01] dB on `fr2_xyz`/`fr1_xyz`, despite maps of 71k/79k. Capping the error-only map at 23k adds +2.76 [2.62, 2.90] and +1.00 [0.96, 1.05] dB, but multiplies depth L1 by 2.8 on `fr1_xyz`. A 39k cap, close to that scene's coverage map size, adds +0.04 [−0.03, 0.10] dB there (and +0.80 [0.67, 0.93] dB on `fr2_xyz`). Thus a static size limit does not reproduce coverage throttling's effect on `fr1_xyz`. For raw-error selection on `fr2_xyz`, coverage reduces the share of modeled update budget spent on Gaussians younger than ten frames from 22.7% to 9.5%, while the per-frame selection rate among mature Gaussians rises from 37.9% to 87.4%. The corresponding mature selection rate on `fr1_xyz` rises from 28.8% to 57.4%. Allocation shifts and the budget intervention support compute dilution under the tested protocol, but coverage throttling also changes the optimization trajectory (its spawn cap carries most of the effect; see the decomposition below). Map size alone is not identified as the unique cause.

Replica factorial and proportional budget (Tab. S5). We ran the same selector×throttle cells on two Replica scenes. Unlike TUM, coverage throttling lowers raw-error PSNR (−0.68 and −0.87 dB on `office0` and `office1`) and shrinks the map (113k to 65k and 84k to 40k); backlog throttling is worse still (−2.55 and −2.04 dB relative to coverage alone, with depth L1 up by 0.12 and 0.05 m). Without throttling, the temporal-influence selector trails raw error by 1.95 and 2.33 dB. Budget interventions help: a fixed ×4 budget adds +2.81 and +1.56 dB, and a budget proportional to map size (mean scale 3.67× and 3.87×, map size unchanged) adds +2.88 and +1.56 dB. On TUM the proportional budget adds +3.73 and +2.99 dB on `fr2_xyz` and `fr1_xyz` over the unthrottled baseline measured in the same batch (24.16 and 21.54 dB). These batches use the same code revision as Tabs. S3 and S4.

Scene-level evidence (Tab. S6). Because the unit of generalization is the scene, we ran the unthrottled, coverage-throttled and map-proportional raw-error policies on 11 further scenes (six Replica, five TUM; 99 frames, three seeds) and pooled them with the four scenes above. The map-proportional budget improves PSNR on all 15 scenes

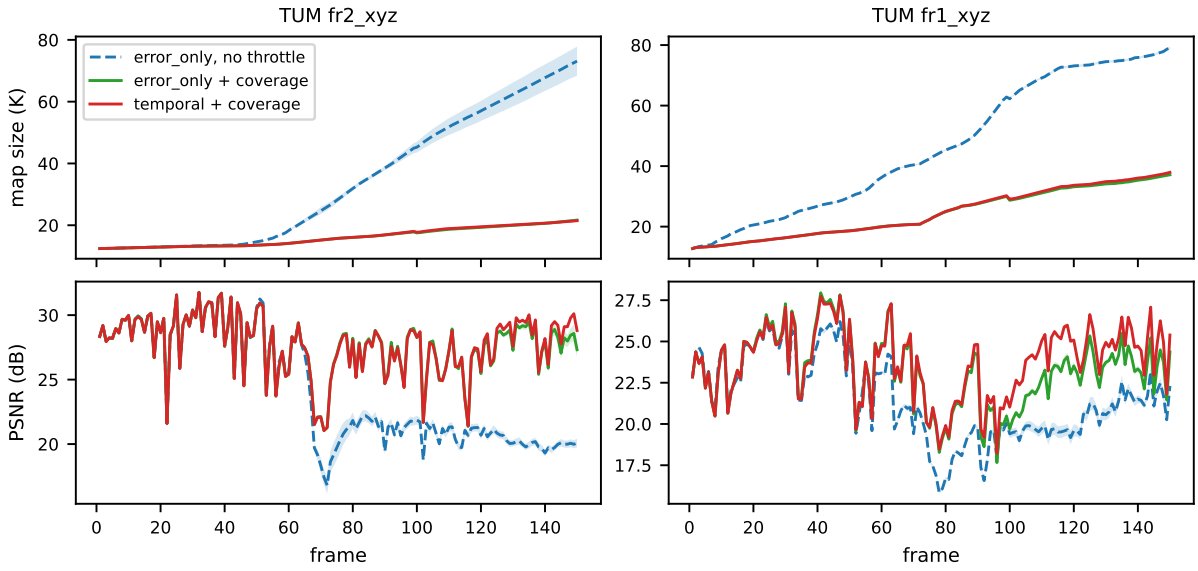


Figure S2. **Map growth and reconstruction quality over time** on TUM `fr2_xyz` and `fr1_xyz` (150 frames, eight seeds). Lines show the seed mean and shaded bands show ±1 standard deviation. `error_only` versus `error + coverage` holds the selector fixed to show the coverage effect; `temporal + coverage` adds the combined policy. Paired sequence-level contrasts are reported in Tab. S4.

	Replica `office0`			Replica `office1`		
Condition	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	30.63	.0403	112.7	33.84	.0161	84.2
Error, coverage	29.95	.0556	65.2	32.97	.0273	39.9
Error, coverage + backlog	27.40	.1720	45.7	30.93	.0744	25.9
Temporal×influence, no throttle	28.68	.0489	115.8	31.51	.0202	85.8
Temporal×influence, coverage	28.14	.0651	66.3	31.73	.0329	42.0
Temporal×influence, coverage + backlog	26.18	.2385	45.8	29.79	.1093	24.1
Error, optimization budget ×4	33.44	.0326	107.5	35.41	.0106	82.8
Error, budget ∝ map size	33.51	.0327	107.5	35.41	.0108	82.7

Table S5. **Replica factorial and map-proportional budget, 99 (`office0`) and 119 (`office1`) frames × 8 seeds per scene.** Same cells as Tab. S4; all runs use the current code revision and a 15 ms modeled budget unless stated. The proportional cell sets the optimization budget to 15 ms × N_t/N_{ref} (mean scale 3.7–3.9×) with the spawn allowance fixed at 15 ms. Coverage and backlog throttling lower PSNR on Replica, whereas more optimization budget raises it.

(median +2.70 dB, range +0.02 to +3.73 dB; exact sign test $p < 10^{-4}$); the one near-zero gain is the small-map static scene `fr3_sitting_static` (+0.02 dB, 17k Gaussians). Coverage throttling improves PSNR on 5 of 15 scenes (median −0.13 dB, range −1.64 to +3.53 dB; sign test $p = 0.30$). The proportional-budget gain tends to grow with the unthrottled map size (Spearman +0.52, $p = 0.048$, $n = 15$, exploratory), as dilution predicts. Per-scene intervals are over three or eight seeds on a fixed trajectory, so the sign and scale across scenes are the evidence. More budget helps any compute-limited optimizer: this shows that raw-error selection under a 15 ms budget is compute limited in every scene tested, not that growth is the cause.

Further diagnostics (Tabs. S8–S11). A budget dose-response (×1/ × 2/ × 4, free or capped map) shows PSNR rising monotonically with the budget on all four scenes, with the fraction of mature Gaussians updated per frame tracking it (Spearman +1.00 over the free-growth cells); caps remove capacity on Replica. On three 500-frame TUM sequences the unthrottled baseline degrades with horizon and the gain from the proportional budget grows from +1.42 to +3.65 dB over frames 1–150 to +5.77 to +8.48 dB over frames 301–500. On held-out frames (every eighth frame rendered, never mapped) the proportional-budget gains keep their sign but retain only 47–76% of their size. With poses from a map-independent tracker, PSNR at the tracked pose improves by +3.12 to +4.48 dB but rendering from the ground-truth pose removes almost all of it. **Decomposing coverage throttling (Tab. S7).** Coverage throttling combines a densification-threshold multiplier (up to 1.5×) with

Scene	Prop.	Cov.	Map (k)	Scene	Prop.	Cov.	Map (k)
fr2_xyz (T)	+3.73	+3.53	75	office1 (R)	+1.56	−0.87	84
fr1_xyz (T)	+2.99	+1.71	79	office2 (R)	+2.76	−0.08	115
fr1_desk (T)	+0.92	−0.16	60	office3 (R)	+3.55	−0.26	148
fr1_room (T)	+1.74	−0.13	103	office4 (R)	+3.68	−0.04	135
fr1_rpy (T)	+2.06	+0.04	72	room0 (R)	+2.54	−0.19	149
fr3_sitting_static (T)	+0.02	+0.08	17	room1 (R)	+2.12	−0.25	102
fr3_structure_texture_near (T)	+2.70	+2.54	61	room2 (R)	+3.09	−1.64	145
office0 (R)	+2.88	−0.68	113				

Table S6. **Scene-level change in mean PSNR (dB) relative to unthrottled raw-error selection, 15 scenes.** Prop.: optimization budget proportional to map size; Cov.: coverage throttling; Map: final size of the unthrottled map. T: TUM, R: Replica. fr2_xyz and fr1_xyz use 150 frames, office0 and office1 use 99 and 119 frames, all with eight seeds; the others use 99 frames and three seeds. Proportional budget is positive on 15 of 15 scenes (median $+2.70$, exact sign test $p < 10^{-4}$); coverage throttling on 5 of 15 (median -0.13, $p = 0.30$). Spearman correlation between map size and proportional-budget gain: $+0.52$ ($p = 0.048$, exploratory).

Condition	fr2_xyz	fr1_xyz	office0	office1
Coverage throttling (full)	+3.36	+1.70	−0.67	−0.88
spawn cap only	+3.25	+1.55	−0.67	−0.91
threshold only	+1.12	+0.18	+0.02	−0.01
Spawn cap ×0.2, ungated	+3.30	+1.54	−0.72	−1.14
Spawn cap ×0.5, ungated	+1.56	+0.54	−0.11	−0.11
Budget ∝ map size	+3.69	+2.96	+2.89	+1.54
Throttling + budget ∝ map	+3.67	+2.82	+1.32	−0.08

Table S7. **Decomposition of coverage throttling, change in mean PSNR (dB) relative to unthrottled raw-error selection, four seeds.** Spawn-cap-only fixes the densification-threshold multiplier at 1; threshold-only disables the spawn cap; the ungated caps apply whenever the view is settled. Budget ∝ map size uses a mean 3.7–6× larger optimization budget than the other rows.

a spawn cap of 0.2× once coverage saturates; four seeds per scene separate them. The spawn cap alone reproduces nearly all of the effect (+3.25 vs +3.36 dB on fr2_xyz, +1.55 vs +1.70 on fr1_xyz, −0.67 vs −0.67 on office0), whereas the threshold multiplier alone gives +1.12, +0.18 and +0.02 dB. A 0.2× spawn cap applied whenever the view is settled, without the coverage gate, matches the full policy on TUM (+3.30, +1.54 dB) and is slightly worse on Replica (−0.72, −1.14 dB); a 0.5× cap gives intermediate results (+1.56, +0.54, −0.11, −0.11 dB). The coverage gate therefore adds little beyond limiting the growth rate. Adding throttling to the map-proportional budget adds nothing on TUM (+3.67 vs +3.69 dB on fr2_xyz) and costs about 1.6 dB on both Replica scenes relative to the proportional budget alone (+1.32 vs +2.89 and −0.08 vs +1.54 dB). The proportional budget spends several times more compute than the fixed-budget rows, so the two controls address different constraints: capping growth when compute is fixed, scaling the budget when it can grow.

Quality at measured cost (Tab. S8). Quality per unit of compute differs from quality per modeled budget, because the map size changes the rendering cost. We sweep the modeled budget (×1, ×2, ×4) with free growth or a ×0.2 spawn cap (four seeds). The proportional-budget gains cost real time: the mean end-to-end frame time rises from 86 to 135 ms on office0. On the TUM scenes the capped map is both faster and as good or better: on fr2_xyz the capped ×1 cell reaches 27.53 dB in 79 ms, versus 24.08 dB in 88 ms for free growth and 27.81 dB in 96 ms for free growth at ×4; on fr1_xyz the capped ×2 cell reaches 24.40 dB in 87 ms versus 24.45 dB in 119 ms for free growth at ×4. On Replica, free growth with more budget is better at equal time: interpolating its curve at the capped cells' frame time gives 0.36–0.64 dB (office0) and 0.96–1.35 dB (office1) more PSNR than the capped ×2 and ×4 cells. Timings come from one laptop GPU and indicate relative costs only. A spawn cap is thus the better use of compute on TUM and the worse on Replica, matching the modeled-budget result, and the proportional budget's gains are paid for in wall-clock time.

Observed failure cases. Two TUM scenes show where the combined policy trails raw-error selection. *Exploration-heavy motion* (fr1_room, −1.04 dB) coincides with a smaller final map (92k vs 97k); the available run does not isolate whether throttling or selection causes the gap. *Rapid rotation* (fr1_rpy, −1.13 dB mean, finals 24.5 vs 24.1) is consistent with a lag from temporal smoothing, but this attribution has not been tested independently. **Horizon effect (Fig. S2):** at 30 frames OURS ties error_only (−0.01 dB on fr2_xyz); the gap appears at longer horizons. The new age-group logs in Tab. S4 link this divergence to a smaller update share for mature Gaussians in the growing map. Around frame 65 in seed 42, both error-only variants encounter difficult views; the unthrottled run continues spawning roughly 540–720 splats per frame and its PSNR remains low after frame 70, whereas the coverage run spawns roughly 60–130 (mean 92) and recovers by frame 75. The unthrottled loss rises from about 0.07 at frame 65 to 0.13 at frame 70 and stays high; opacity declines while the map expands. This is sustained divergence following shared hard views, not

Scene	Free map growth ×1	Free map growth ×2	Free map growth ×4	Spawn cap ×0.2 ×1	Spawn cap ×0.2 ×2	Spawn cap ×0.2 ×4
fr2_xyz	24.08 (88)	26.50 (96)	27.81 (96)	27.53 (79)	27.92 (62)	27.93 (63)
fr1_xyz	21.51 (100)	22.77 (108)	24.45 (119)	23.10 (92)	24.40 (87)	24.44 (76)
office0	30.62 (86)	32.26 (105)	33.44 (135)	29.89 (79)	31.31 (98)	31.77 (109)
office1	33.85 (86)	34.86 (107)	35.42 (138)	32.74 (80)	33.48 (98)	33.53 (107)

Table S8. **Quality at measured cost.** Mean PSNR (dB) with the mean end-to-end frame time in ms in parentheses, for a modeled optimization budget of $15 \times \{1, 2, 4\}$ ms (spawn allowance fixed) with free growth or a $\times 0.2$ spawn cap; raw-error selection, four seeds. Timings come from one laptop GPU and indicate relative costs only.

evidence for a single isolated gradient explosion. The logs cannot rule out floaters without a spatial geometry audit.

Scene dependence (Tab. S7). The combined policy wins the two confirmatory TUM comparisons ($n = 8$): fr2_xyz +3.53 dB and fr1_xyz +2.38 dB over raw-error selection, both $p = 0.0078$. It trails raw-error selection on all eight Replica scenes (-1.93 to -4.21 dB; office0 and office1 at $n = 8$, the rest exploratory at $n = 3$) and on fr1_room and fr1_rpy, so across 14 sequences it loses mean PSNR on 10. On Replica office0, raw-error selection with a weak throttle ($\kappa = 0.98$) reaches 30.17 dB versus 26.85 dB for the combined policy (supplement). The Replica factorial (Tab. S5) locates the loss: the unthrottled raw-error cell is the best of the six factorial cells on both scenes, adding the temporal-influence selector costs 1.95 and 2.33 dB, and coverage throttling costs 0.68 and 0.87 dB for raw-error selection. We cannot tell whether this reflects Replica's rendering, trajectories or depth quality. Compute allocation still matters there, since a map-proportional budget adds +2.88 and +1.56 dB. A simple hole-rate router that picks the better arm on the two datasets is in the supplement; its within-scene depth interventions did not reverse the ranking, so we do not use it as evidence for any mechanism.

Systems. Scheduler overhead is 2.54 ms; optimization-budgeted execution runs at about 10 FPS, so this is *not* end-to-end 15 ms real time. A separate baseline odometer (dense photometric+depth Gauss-Newton, no keyframes or loop closure) reaches ATE 0.9/2.0 cm on fr2_xyz/fr1_xyz over 100 frames (Tab. S3); it is a sanity check, not a SLAM claim. AlexNet LPIPS on fr2_xyz (150f×3) is 0.211 for OURS versus 0.294 for error_only (earlier code revision); depth L1 is in Tab. S4, and we report no mesh-based surface accuracy. Published-system numbers use different frames, resolution and views and give range context only (Tab. S2); short-horizon budget-curve and coverage-knee ablations are in Figs. S1–S2. **Reproducibility.** Every table maps to a frozen bundle and a command in paper_repro/; the headline bundle records per-policy configs, source hashes, artifact hashes, pose protocol and clean-worktree status of each run set. Prior pose-bugged numbers (19.24 dB) are quarantined in results/final_confirmation/DEPRECATED.md and never compared. Environment: RTX 4050 Laptop GPU (6 GB), PyTorch 2.11+cu128, gsplat [**?**] 1.5.3, Python 3.12; peak allocation stays under 3.5 GB with the desktop compositor active, which is the intended deployment envelope for budgeted online mapping.

B.3. Time-Aware Control

The diagnosis suggests two controllable quantities. **Time-targeted budget (TT).** The modeled optimization budget ignores the rendering time that grows with the map, so a compact map leaves wall-clock time unused (Tab. S8). TT keeps the measured end-to-end frame time near a target T by scaling the modeled budget with a multiplier $m \in [1, m_{\max}]$:

$$m \leftarrow \mathrm{clip}\big(m\,(T/\hat{\tau})^{\gamma},\, 1,\, m_{\max}\big), \tag{2}$$

where $\hat{\tau}$ is an exponential moving average of the measured frame time, including the controllers' own overhead. The spawn allowance stays at the base budget, so extra compute cannot buy extra splats. **Value-gated growth (VGG).** Every second frame, before the incoming frame is used for any update, we render it twice with the current map: once as is and once with the Gaussians created in the last five frames made transparent. The PSNR difference per 1000 recently spawned Gaussians, v, is the counterfactual value of those Gaussians (with the updates they received) on an unseen view. An exponential moving average of v drives a multiplicative rule with hysteresis on the spawn-cap factor $g \in [0.1, 1]$: $g \leftarrow 0.25\,g$ if the average is below $\theta_{\mathrm{lo}} = 0.08$ dB/kG and $g \leftarrow 2g$ if it is above $\theta_{\mathrm{hi}} = 0.20$ dB/kG. When growth is clearly valuable ($> 2\theta_{\mathrm{hi}}$) the measurement interval doubles up to 16 frames, which removes most of the overhead on scenes where growth always pays. **Constants and protocol.** $T = 100$ ms, $\gamma = 0.5$, $m_{\max} = 16$, the VGG constants above, an average coefficient of 0.7 and a minimum of 200 recently spawned Gaussians per measurement were chosen on four development scenes (fr1_desk, fr3_structure_texture_near, replica_room1, replica_office2; one or two seeds) and frozen before any evaluation scene was run. Evaluation uses the other 11 scenes and five fresh 150-frame windows (frames 200–450 of the long TUM sequences), none of which was used to set a constant. Higher VGG

Method	11 held-out scenes > 0	worst	mean	Δms	5 fresh windows > 0	mean	Δms
Fixed spawn cap ×0.2	4/11	−1.67	+0.04	-8	3/5	+0.55	-8
Budget ∝ map (+38 ms)	11/11	+0.05	+2.54	+38	5/5	+1.18	-0
VGG alone	6/11	−0.08	+0.25	+2			
TT alone	11/11	+0.06	+1.22	+5	5/5	+0.77	-5
TT + fixed cap	9/11	−0.55	+1.03	-1	5/5	+0.87	-22
TT + VGG (ours)	11/11	+0.05	+1.29	+4	5/5	+0.90	-5

Table S9. **Controllers versus the unthrottled baseline (modeled 15 ms budget), change in mean PSNR (dB).** Left: 11 scenes not used to set any constant (hyperparameters were frozen on four development scenes); right: five fresh 150-frame windows of long TUM sequences. > 0: scenes improved; worst/mean over scenes; Δms: mean change in end-to-end frame time. TT: time-targeted budget ($T = 100$ ms); VGG: value-gated growth. Seed-paired, three or four seeds per scene.

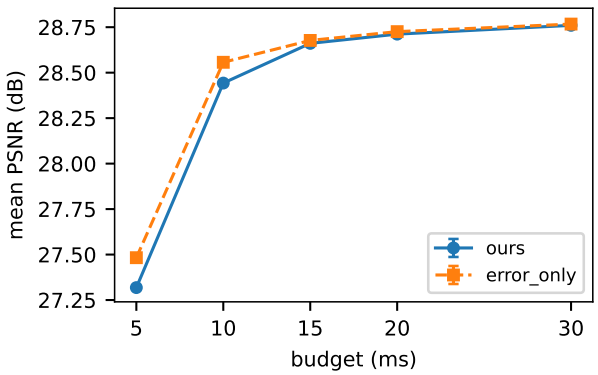


Figure S3. **Budget curve** (fr2_xyz, 30 frames×3, matched modeled budgets): policies are similar across the tested budget range at this short horizon. This experiment does not establish where the 150-frame policies separate as a function of budget.

thresholds did not help on the development scenes and were not adopted.

B.4. Evaluating the Controllers

Controllers versus the baseline (Tab. S9). TT+VGG improves PSNR on all 11 held-out scenes (median +1.14 dB, mean +1.29 dB, worst +0.05 dB) with a mean frame time only 4 ms above the baseline's, and on all five fresh windows (mean +0.90 dB, 5 ms faster). A fixed spawn cap, the usual remedy, helps on only 4 of 11 scenes and loses up to 1.67 dB; the map-proportional budget gains more (+2.54 dB) but spends 38 ms more per frame. On 8 of 11 scenes TT+VGG is both better than the baseline and within 15% of its frame time. Per-scene results are in Tabs. S14–S15. **Contribution of each controller.** Most of the gain comes from TT (mean +1.22 dB, positive on all 11 scenes). VGG adds little on average (+0.06 dB on the 11 scenes, positive on 3, worst −0.07 dB; +0.14 dB on the fresh windows, positive on all five), but it matters where growth is wasteful: on `fr2_xyz` it raises the gain from +2.94 to +3.88 dB while cutting the frame time from 84 to 72 ms. It does not reproduce the losses of a fixed cap on Replica: TT with a fixed cap loses up to 0.55 dB against the baseline, whereas the worst case of TT+VGG is +0.05 dB. VGG is conservative: on `fr1_xyz` TT with a fixed cap reaches +2.90 dB against +1.14 dB for TT+VGG. A hard cap on the number of Gaussians equal to TT+VGG's own final map size, the control behind count-budgeted densification, averages −0.25 dB against +1.28 dB for TT+VGG, which wins on 10 of 11 scenes (Tab. S13). **Equal measured time (Tab. S16).** Because TT turns a latency target into a budget, we sweep $T \in \{70, 85, 100, 120, 140\}$ ms and compare with free growth at fixed modeled budgets $\times 1/\times 2/\times 4$. On `fr2_xyz`, TT+VGG reaches 27.94 dB in 72 ms, versus 24.08 dB in 88 ms for the baseline and 27.81 dB in 96 ms at ×4; on `fr1_xyz` at $T = 120$ ms it reaches 23.93 dB in 102 ms versus 21.92 dB for the fixed-budget curve interpolated at that time. On Replica it matches the best fixed multiplier at equal time (+0.01 to +0.23 dB), so there the benefit is that a latency target replaces a hand-tuned multiplier. **Held-out views.** On frames that are rendered but never mapped (every eighth frame), TT+VGG's gains stay positive but shrink: +2.15, +1.16, +0.69 and +0.34 dB on `fr2_xyz`, `fr1_xyz`, `office0` and `office1`, which is 79%, 76%, 55% and 45% of the gains on mapped frames. The held-out frames lie on the mapped trajectory, so this proxy does not test generalization to new trajectories. **Where the gains do not hold.** The controllers rely on the budget being the bottleneck, which holds on every scene tested but with a tiny effect on the small, static `fr3_sitting_static` ($\leq$ +0.09 dB). VGG measures the value of recent spawns on one incoming frame and ignores their long-run cost, which is why it stays conservative on some TUM scenes. The estimated-pose check (Tab. S11) shows that gains in self-consistent quality vanish when rendering from the true pose, so the improvement is bounded by pose accuracy.

C. Short-horizon ablations

D. Historical and Cross-System Context

The following tables use different substrates or protocols from the corrected matched-budget benchmark. **Historical component ablation (Tab. S10, pose-bugged substrate, 150 frames, 5 seeds).** Relative to A0, A2 (coverage) differs by +1.04 dB and A3 (coverage plus backlog) by +1.08 dB. The direct A3–A2 increment is +0.04 dB; no direct significance test is reported, and comparisons against A0 have Holm-adjusted $p = 0.25$. These numbers are not transferred to the corrected benchmark. A native local SplaTAM run on the same sequence (144f, seed 42: final 27.20 / mean 23.03 dB, SSIM 0.974, LPIPS 0.046, depth 1.26 cm, ATE

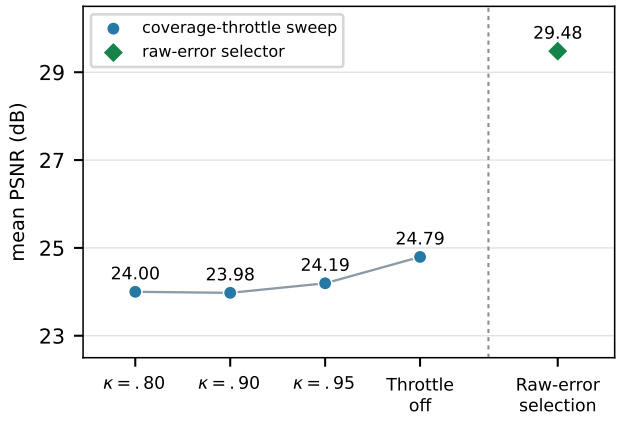


Figure S4. **Replica office0 ablation** (30 frames, $n = 3$). Disabling coverage throttling changes mean PSNR by 0.82 dB; switching from temporal×influence to raw-error changes it by a further 4.69 dB. These are descriptive short-run differences without a reported significance test.

Condition	Mean	Δ vs A0	Map	Finding
A0 baseline	16.07	—	140,998	bloated
A1 + warmup	15.32	−0.75	140,038	harmful
A2 + coverage	17.11	+1.04	44,865	ns vs A0
A3 + backlog	17.15	+1.08	43,824	+0.04 vs A2
A4 + learned	14.53	−1.54	38,317	harmful

Table S10. **Historical component ablation on the pose-bugged substrate** (5 seeds). Values and tests are not evidence for effects on the corrected benchmark. `results/step1_ablation_5seeds/`.

0.41 cm) uses a different resolution and tracking setup; these figures are contextual only and do not test compute dilution.

E. Additional Evaluation

F. Further Experiments

The following experiments would refine the mechanism and extend its scope. **Map-growth mediation:** vary continuous spawn rates and pruning strength while holding the candidate error threshold fixed; compare trajectories at matched map size and depth quality. A budget proportional to current map size would also test whether the fourfold rescue persists under a less coarse intervention. **Router generalization:** obtain a third sensor domain and scenes outside the reported set for threshold calibration; then test structured missing-depth artifacts and geometry-preserving TUM hole repair. The present random-hole and valid-depth-smoothing interventions did not reverse the preferred policy. **Longer mapping evaluation:** use sequences of at least 500 frames, multiple start windows, a real tracker, and depth or geometric metrics. **Corrected utility labels:** regenerate counterfactual labels on the corrected pose substrate before drawing conclusions about learned ranking. **Finer-grained adaptivity:** the 7-scene routing table shows motion/exploration regime matters within a sensor (fr1_room, fr1_rpy trail by ≈ -1.1 dB); extending the observable cue beyond hole-rate (e.g. optical-flow magnitude, coverage velocity) could route rapid-motion frames to instantaneous error while keeping temporal smoothing elsewhere.

G. Hole-rate policy routing (auxiliary)

Tables S4–S6 come from earlier code revisions, before the evaluator correction (standard windowed SSIM; alpha-normalized depth). Their SSIM and depth values are not comparable with the main-paper tables, and their PSNR may differ by a few tenths of a dB. Main-paper scene dependence motivates asking whether an observable cue can pick the better policy. We report this as an auxiliary analysis; it does not support a mechanism claim. **Hole-rate policy routing.** Raw depth hole-rate separates the evaluated TUM and Replica datasets: TUM 0.18–0.31 vs Replica ≈ 0.000 (300–500×). Since the datasets also differ in rendering, pose quality, and scene content, this association does not establish that sensor noise causes the policy ranking to change. The score $n = \text{clip}(2.5 \cdot \text{holes} + 1.0 \cdot \text{edge})$ routes $n \leq 0.10 \rightarrow$ raw-error selection ($\kappa = 0.98$) and $n \geq 0.30 \rightarrow$ temporal×influence + throttling ($\kappa = 0.90$), computed pre-hygiene in milliseconds on CPU. Formally, with h_t the raw missing-depth rate and e_t the raw edge density of frame t:

$$n_t = \min(1,\, 2.5h_t + 1.0e_t), \quad \pi_t = \begin{cases} \texttt{err_only} & n_t \leq 0.10 \\ \text{blend} & 0.10 < n_t < 0.30 \\ \texttt{temporal} & n_t \geq 0.30 \end{cases} \tag{3}$$

with coverage knee $\kappa \in \{0.98, 0.90\}$ tied to the chosen arm. The score and thresholds were chosen after inspecting five development scenes (Fig. S5), including `fr1_xyz`. That scene is therefore development evidence for the router, although its separate 150-frame mapping-policy comparison uses frozen settings. A seven-scene audit is reported separately in Tab. S15; scene-level independence must be tracked separately from seed-level repeats. Two earlier features failed held-out separation and are reported as such: post-hygiene invalid ratio (conflates far-but-valid geometry — 18% of Replica room0 exceeds 4 m — with holes) and Laplacian texture energy (measures sharpness, inverted: crisp synthetic scores *higher*). Operating on pre-hygiene raw depth was the fix (Fig. S5). A mixed-regime test injects 30% holes mid-trajectory on Replica and the router changes arms on 49/50 frames at the transition. This verifies feature-triggered switching in that test; end-to-end PSNR trails fixed `error_only` by 0.85 dB, so a quality benefit on mixed real sequences is not established.

Policy choice differs across the two evaluated substrates (Tab. S13). On synthetic Replica the fixed combined policy loses 3.32 dB; routing to raw-error selection recovers 3.32 [3.29, 3.35], $p = 0.0078$. **Seven-scene routing (Tab. S15, exploratory $n = 3$, no scene labels):**

Method	Replica	fr2_xyz	Protocol
MonoGS	37.50	16.17	full seq + refine
Point-SLAM	35.17	—	full seq + refine
RTG-SLAM	32.79	17.08	full seq + refine
SplaTAM	—	24.50	full seq + refine
SplaTAM (local run, 144f)	—	27.20 final / 23.03 mean	SLAM 640×480, no refine
Photo-SLAM	—	21.07	full seq + refine
Ours (error_only)	20–33	24.21	budgeted, 100–150f
Ours (throttled)	20–29	**27.74**	budgeted, 100–150f

Table S11. **Published-SOTA context (protocol-mismatched) and one local SplaTAM run.** Published results are shown for context, not as matched comparisons. The local run uses the same sequence but a different resolution and tracking setup; details are in the text.

Method	fr2_xyz	fr1_xyz	Horizon	System?
Ours (GN VO)	0.9	2.0	100f	no (baseline)
MonoGS	1.71	—	full	yes
SplaTAM	0.50	—	full	yes
Photo-SLAM	0.63	—	full	yes

Table S12. **ATE RMSE (cm).** Rows 2–4 are published full-system numbers; ours is a decoupled odometry sanity check on fixed horizons.

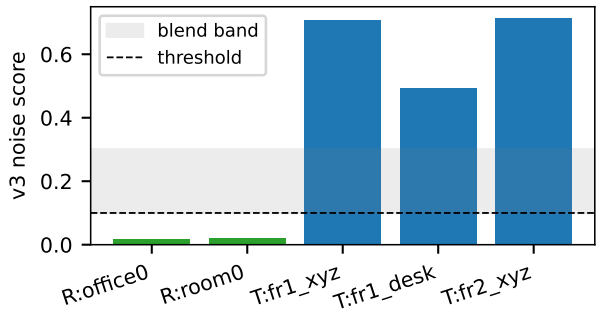


Figure S5. **Depth score on five development scenes.** The five plotted scenes include `fr1_xyz`; its mapping-policy comparison is distinct from independent router validation. The seven-scene router audit appears in Tab. S15. The separation shown here measures dataset association, not sensor-noise causality.

Policy	Mean	Final	SSIM	Map
Fixed OURS	26.85	26.07	0.9244	78,825
Routed (err)	30.17	28.02	0.9584	115,964
Δ	+3.32 ✓	+1.95 ✓	+0.034	—

Table S13. **Adaptive validation, Replica `office0`, 99 frames × 8 seeds.** Router picks `error_only` (holes ≈ 0.02): $p = 0.0078$, $d_z = 71.9$. No fixed policy wins both substrates.

routed PSNR is close to fixed-policy reference values on five scenes and about 1.1 dB lower on exploration-heavy `fr1_room` and rapid-rotation `fr1_rpy`. These are unpaired, exploratory comparisons; the table reports references

Scene	Level	Error	Temporal	Δ
Replica office0	0%	30.18	26.84	−3.33
	15%	30.32	26.75	−3.58
	30%	30.31	26.62	−3.69
TUM fr2_xyz	0	24.12	27.69	+3.57
	0.5	24.10	27.64	+3.54
	1.0	24.21	27.62	+3.41

Table S14. **Within-scene depth interventions** (150 frames on TUM, 99 on Replica, × 3 seeds). Δ is temporal×influence with coverage minus unthrottled raw error, measured on identical perturbed frames. Replica also receives Gaussian depth noise with standard deviation 0.05 times the hole fraction in meters. TUM smoothing acts only on valid depth after the standard hygiene stage; it does not remove remaining holes. No level reverses policy ordering. Full paired intervals: `results/depth_intervention/analysis.json`.

Scene	Routed PSNR	Fixed reference	Route
fr2_xyz 150f	27.66	27.64 (ours)	temporal
office0 99f	30.36	30.17 (err)	error_only
office1 100f	33.11	33.48 (err)	error_only
room1 100f	24.17	24.29 (err)	error_only
fr3_static 100f	25.95	25.96 (ours)	temporal
fr1_room 100f	15.50	16.54 (err)	temporal
fr1_rpy 100f	19.93	21.06 (err)	temporal

Table S15. **7-scene router audit.** Routed PSNR uses $n = 3$; fixed values are unpaired references. Office0 uses $n = 3$ routed vs $n = 8$ fixed. Five scenes are held out; two overlap development.

rather than inferential deltas. Domain dependence is thus finer than real-vs-synthetic: motion and exploration regime matter within a sensor, and the current router does not capture them — reported as-is. **Depth intervention (Tab. S14).** On Replica `office0`, raising random missing-depth rate to 30% with accompanying 1.5 cm Gaussian noise makes the frozen score exceed its temporal-route threshold, yet raw-error selection remains ahead by 3.69 [3.62, 3.78] dB across three seeds. On TUM `fr2_xyz`, bilateral smoothing of valid depth after the standard hygiene stage changes the temporal-minus-error advantage only from +3.57 to +3.41 dB. These

interventions do not support a causal claim that missing depth alone determines the better policy. The Replica masks are independent pixel noise rather than a full Kinect artifact model; the TUM intervention leaves missing pixels masked. Thresholds were not recalibrated on independent scenes because every available TUM and Replica scene appears in the reported scene set, and no third sensor domain is present locally.

H. Counterfactual utility and negative results (auxiliary)

Measurement instrument. Per-frame scheduling uses $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ and ΔQ measured by exact full-scene counterfactual renders (active edits embedded in all N splats, avoiding depth-interleaving gradient bias of affine-background approximations). The oracle $U_i^\star = \Delta Q_i^{\text{exact}} / \Delta C_i$ is our measurement instrument, never a runtime input. It reveals: (i) negative utility in 2.7–38.7% of interventions by stratum (pose-fixed regeneration), with only 2/75 negative interventions at depth discontinuities; (ii) interaction is protocol-dependent — corrected joint-group measurements give $R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$ (relatively additive), while a separate pairwise protocol shows stronger interaction; pointwise top-K is therefore treated as a reference, not a bound, without a universality claim; (iii) offline TwoHeadMLP rank signal ($\rho \approx 0.20$, NDCG@20 0.46, measured on pre-pose-fix labels whose absolute values are polluted but whose within-set method comparison remains valid) reverses to -2.51 dB closed-loop. Conditional-oracle selection numbers are withheld pending a rerun under fork-correct semantics. Pose-fixed counterfactuals are negative in 29/75 flat, 16/75 texture, 14/75 edge and 2/75 depth-discontinuity interventions (candidate-level proportions from five seeds; no cluster-aware intervals). Uniform random selection from the same stratified pools has an expected negative-intervention rate of 20.3% (95% seed-cluster bootstrap CI 16.0–24.3%), so the stratum range alone does not establish ranking quality; the per-candidate labels needed to compare the selected top-error subset were not saved. **Negative results. Learned utility reverses closed-loop** (-2.51 dB in the historical one-seed factorial on the pre-fix substrate; GradNorm beats the MLP pointwise $0.51 > 0.25$): offline labels ignore trajectory shift and inter-Gaussian coupling, so we ship no learned component. This result is separate from the corrected-pose factorial in Tab. 2 of the main paper. **Contextual re-ranking buys no online gain**: adaptive greedy shows no significant improvement over static pointwise selection in direct multi-seed budget sweeps (Wilcoxon n.s., CIs span zero) while incurring 424× selection overhead (76.36 vs 0.18 ms). Supporting rank-stability/oracle-gap numbers predate the fork-semantics fix and are withheld pending regen.

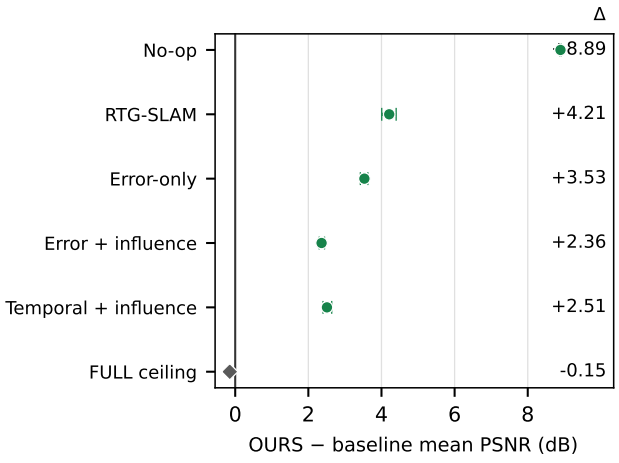


Figure S6. **Paired mean-PSNR differences** on fr2_xyz (150 frames, $n = 8$). Points show OURS − baseline; bars are 95% paired-bootstrap CIs. Positive favors OURS. FULL is 0.15 dB higher; all six contrasts pass Holm correction ($p = 0.0469$).

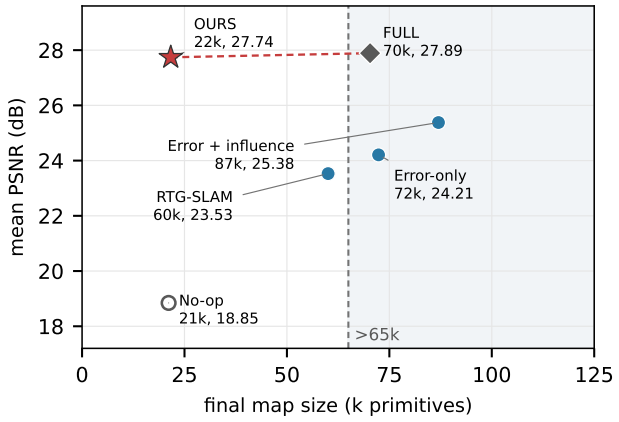


Figure S7. **Quality–map-size trade-off** on fr2_xyz (150 frames, $n = 8$). Among policies that optimize, only OURS and FULL lie on the frontier; FULL gains 0.15 dB with a 3.2× larger map. The four growing baselines hold 60–87k Gaussians; no-op stays compact at lower quality.

I. Additional main-benchmark figures

J. Per-scene breakdown of the combined policy (earlier code revision)

K. Further diagnostics

Dose-response (Tab. S17). With free growth, PSNR rises monotonically with the budget multiplier on all four scenes (for example 24.24, 26.49, 27.82 dB on `fr2_xyz`), and the fraction of mature Gaussians updated per frame rises with it; their per-seed Spearman correlation over the three free-growth cells is +1.00 in every scene. Over all six cells it

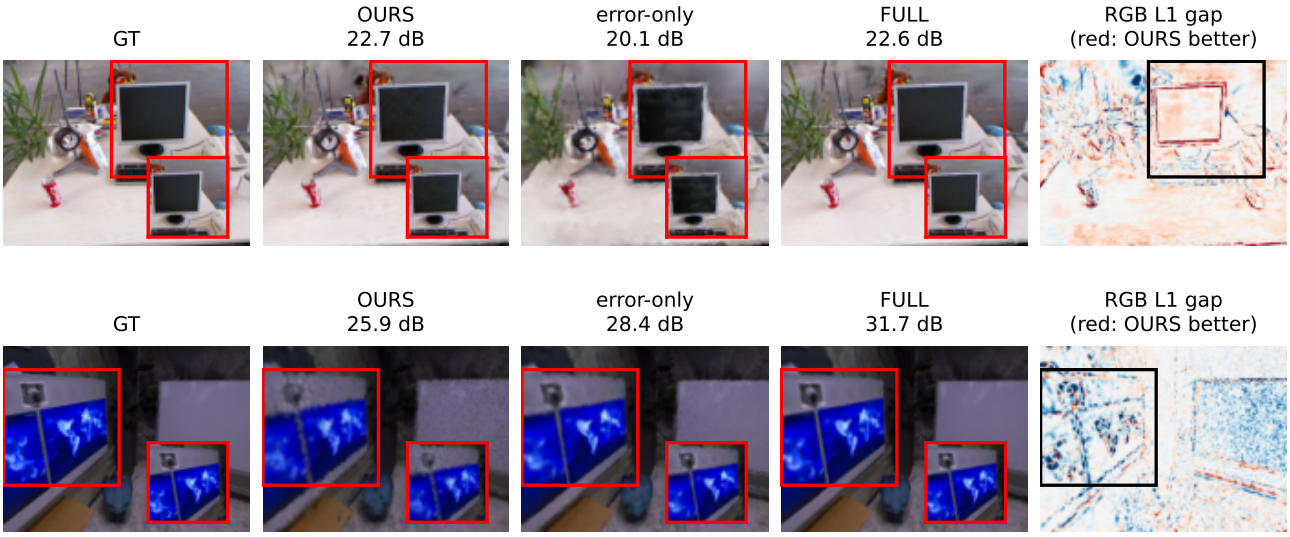


Figure S8. **Qualitative comparison across both evaluated substrates.** Top: an intermediate TUM `fr2_xyz` view (90), where OURS has lower per-view error than `error_only`; bottom: Replica `office0` view 98, where the ranking reverses. Columns show ground truth, OURS, `error_only`, FULL, and signed RGB L_1 error gap (red: OURS has lower error). Insets mark the same crop in each row. This figure comes from an earlier code revision. PSNR labels are per-view values from seed 42, shown only to illustrate the visual comparison; they are not sequence means.

Scene	Frames	OURS	`error_only`	Δ	Map (k)	n
fr2_xyz (TUM)	150	27.74	24.21	+3.53 ✓	22/ 72	8
fr1_xyz (TUM)	150	23.47	21.09	+2.38 ✓	39 / 81	8
fr1_desk (TUM)	150	22.55	22.34	+0.21	27 / 56	3
fr1_room (TUM)	100	15.53	16.54	−1.01	92 / 97	3
fr1_rpy (TUM)	100	19.92	21.06	−1.14	45 / 62	3
fr3_static (TUM)	100	25.96	25.88	+0.08	13 / 18	3
office0 (Rep.)	99	26.85	30.17	−3.32 ✓	79 / 116	8
office1 (Rep.)	100	29.25	33.46	−4.21 ✓	42 / 74	8
office2 (Rep.)	99	22.08	24.67	−2.59	70 / 117	3
office3 (Rep.)	99	22.45	24.47	−2.02	77 / 150	3
office4 (Rep.)	99	23.58	25.93	−2.36	81 / 141	3
room0 (Rep.)	99	20.85	22.78	−1.93	82 / 150	3
room1 (Rep.)	100	20.42	24.29	−3.87	47 / 59	3
room2 (Rep.)	99	22.86	25.88	−3.03	78 / 146	3

Table S16. **Per-scene combined-policy breakdown, 14 sequences.** ✓ marks the reported significant comparisons at $n = 8$; other entries are exploratory. The combined policy wins the two confirmatory TUM comparisons but loses on both confirmatory Replica comparisons; remaining scenes have exploratory sample sizes.

stays $+0.92$ and $+0.75$ on the TUM scenes, where a 23k cap is close to the natural map size, but not on Replica ($+0.20$ and -0.26), where the caps (60k, 40k) remove capacity and PSNR falls by 6–11 dB regardless of budget. Under a cap, $\times 2$ and $\times 4$ coincide once coverage reaches 1.00, because the budget exceeds what the capped map can use. Update coverage therefore tracks quality wherever capacity is not binding. These are within-scene interventions on raw-error selection. They show that the system is compute limited and that the limit tightens as the map grows, but more compute improves any compute-limited optimizer, so they do not by themselves show that growth causes the limitation; the cap cells address that only on TUM.

Longer horizon (Tab. S18). On the three TUM scenes whose sequences allow it (`fr2_xyz`, `fr1_xyz`, `fr1_desk`; 500 frames, two seeds), the unthrottled raw-error baseline degrades with horizon (window-mean PSNR 24.22 → 17.88, 21.50 → 19.32 and 22.49 → 16.95 dB

Scene (cap)	Free map growth ×1	Free map growth ×2	Free map growth ×4	Hard map cap ×1	Hard map cap ×2	Hard map cap ×4	ρ_{all}	ρ_{free}
fr2_xyz (23k)	24.24 (0.38)	26.49 (0.58)	27.82 (0.98)	26.90 (0.73)	27.85 (1.00)	27.85 (1.00)	+0.92	+1.00
fr1_xyz (23k)	21.53 (0.29)	22.78 (0.55)	24.45 (0.89)	22.52 (0.65)	23.63 (1.00)	23.63 (1.00)	+0.75	+1.00
office0 (60k)	30.61 (0.17)	32.25 (0.34)	33.43 (0.72)	24.37 (0.24)	25.25 (0.49)	26.07 (1.00)	+0.20	+1.00
office1 (40k)	33.85 (0.18)	34.88 (0.33)	35.42 (0.68)	22.79 (0.38)	22.85 (0.75)	22.90 (1.00)	-0.26	+1.00

Table S17. **Dose-response in optimization budget, 8 seeds.** Raw-error selection, no throttling; entries are mean PSNR (dB) with the mature-update coverage in parentheses, for a budget of $15 \times \{1, 2, 4\}$ ms with the spawn allowance fixed at 15 ms, with free growth or a hard Gaussian cap. ρ is the per-seed Spearman correlation between PSNR and mature-update coverage over the cells (all six, or the three free-growth cells), averaged over seeds. Once coverage reaches 1.00 extra budget is unused and PSNR saturates. On Replica the caps remove needed capacity (PSNR 6–11 dB lower at equal budget), so the cross-regime correlation does not hold there.

Scene	Frames	Base	Cov.	Prop.
fr2_xyz	1–150	24.22	+3.47	+3.65
	151–300	18.70	+6.07	+9.30
	301–500	17.88	+4.85	+8.48
fr1_xyz	1–150	21.50	+1.71	+3.03
	151–300	19.74	+1.88	+5.37
	301–500	19.32	+0.72	+5.77
fr1_desk	1–150	22.49	+0.47	+1.42
	151–300	18.98	+2.52	+5.05
	301–500	16.95	+1.17	+5.92

Table S18. **Long horizon (500 frames, two seeds).** Mean PSNR of the unthrottled raw-error baseline (Base) and the change from coverage throttling (Cov.) and the map-proportional budget (Prop., mean scale 5.4–6.7×) in three frame windows. The baseline degrades with horizon and the gains grow; coverage throttling is not monotone. Replica sequences are too short for this test.

Mapped / held-out	fr2_xyz	fr1_xyz	office0	office1
Coverage throttling	+2.73/+2.13	+1.50/+1.04	−0.62/−0.27	−0.93/−0.57
Spawn cap ×0.2	+2.68/+2.11	+1.38/+0.96	−0.77/−0.50	−1.08/−0.54
Budget ∝ map	+2.85/+2.18	+2.59/+1.69	+2.58/+1.40	+1.44/+0.68
Gap, unthrottled	0.87	0.98	3.29	2.79
Gap, budget ∝ map	1.53	1.87	4.47	3.55

Table S19. **Mapped versus held-out views, change in mean PSNR (dB) relative to unthrottled raw-error selection, four seeds.** Every eighth frame is rendered but never mapped (a same-trajectory proxy, not an independent trajectory). The last rows give the train-minus-held-out PSNR gap.

from frames 1–150 to 301–500) and its final map reaches 116–136k Gaussians. The map-proportional budget (mean scale 5.4–6.7×) recovers +3.65 → +8.48, +3.03 → +5.77 and +1.42 → +5.92 dB over the same windows, whereas coverage throttling is not monotone (+3.47 → +4.85, +1.71 → +0.72, +0.47 → +1.17 dB). The gap to the unthrottled baseline therefore widens with horizon, consistent with dilution accumulating as the map grows, but with two seeds and three scenes this is descriptive, and the proportional budget spends several times more compute at 500 frames. Replica sequences are too short (99–119 frames) for this test.

Held-out views (Tab. S19). Every eighth frame is rendered but never mapped or optimized. The sign of every effect is unchanged on these frames but its size shrinks: the proportional budget's gain is +2.18, +1.69, +1.40 and +0.68 dB held out versus +2.85, +2.59, +2.58 and +1.44 dB on mapped frames (47–76% retained), and the train-minus-held-out gap grows with the budget (for example 3.29 → 4.47 dB on office0). Throttling's held-out changes are +2.13 and +1.04 dB on TUM and −0.27 and −0.57 dB on Replica. Part of the gain from more optimization is therefore specific to the mapped views. The held-out frames lie on the mapped trajectory, so this proxy does not test generalization to new trajectories.

Estimated poses (Tab. S20). We also map with poses from a frame-to-frame dense RGB-D Gauss-Newton tracker (ATE 1.24 and 2.45 cm; it does not use the map, so ATE is shared across conditions). PSNR at the tracked pose improves by +4.09 to +4.44 dB on fr2_xyz and +3.12 to +4.48 dB on fr1_xyz, more than with ground-truth poses. Rendering the same maps from the ground-truth pose removes almost all of this (+0.02 to +0.03 and +0.15 to +0.22 dB; baseline 21.88 and 17.58 dB versus 27.35 and 23.75 dB at the tracked pose), because pose error rather than optimization budget then limits the alignment. Gains in self-consistent quality therefore do not imply better alignment with the true camera. In a map-based SLAM system a better map could also improve tracking, which this map-independent tracker cannot show.

Closed-loop map-based tracking (Tab. S21). To let map quality feed back into tracking, each frame's pose is estimated by rendering the current map and optimizing a 6-DoF correction coarse to fine (render scales $1/4, 1/2, 1$ with $20/15/10$ Adam steps, learning rate 10^{-3}, constant-velocity initialization) on a photometric plus depth L1 loss, as in SplaTAM-style tracking but without keyframes, loop closure or global refinement; the controllers then map with these poses. The settings were explored on the first 60 frames of the development scenes fr1_desk and

Scene	Condition	Tracked pose	GT pose
fr2_xyz (ATE 1.24 cm)	unthrottled	27.35	21.88
	coverage	+4.31	+0.02
	spawn cap ×0.2	+4.09	+0.03
	budget ∝ map	+4.44	+0.02
fr1_xyz (ATE 2.45 cm)	unthrottled	23.75	17.58
	coverage	+3.40	+0.15
	spawn cap ×0.2	+3.12	+0.15
	budget ∝ map	+4.48	+0.22

Table S20. **Mapping with estimated poses, three seeds.** PSNR (dB) of the unthrottled baseline and the change for each condition, evaluated at the tracker's pose and when the same map renders from the ground-truth pose. The tracker is a frame-to-frame RGB-D Gauss-Newton odometer that does not use the map.

Scene	Base ATE	TT+VGG ΔPSNR	TT+VGG ΔATE	Trop. ΔPSNR
fr2_xyz	7.5 cm	+2.83	−1.61 cm	+3.69
fr1_xyz	6.4 cm	+1.89	−0.92 cm	+3.57
fr1_rpy	12.6 cm	+1.42	+1.50 cm	+2.76
fr3_sitting_static	14.0 cm	+0.10	−0.34 cm	+0.14
fr1_room†	96.7 cm	−0.10	+4.69 cm	+1.39

Table S21. **Closed-loop mapping with map-based tracking, three seeds.** Poses come from aligning each frame to the current map (Supp. text); the same controllers then map with the estimated poses. ΔPSNR (dB) is measured at the tracked pose, ΔATE is TT+VGG minus the unthrottled baseline. †Tracking diverged for every condition and the scene is excluded from the summary: over the four converged scenes TT+VGG changes PSNR by +1.56 dB on average and ATE by a median of −0.63 cm (largest |Δ| 1.61 cm).

fr3_structure_texture_near, not on test scenes; ATE there was 3.6–6.5 cm across the handful of settings tried (single runs, noisy), and we kept the cheaper 20/15/10 schedule at 4.4 cm rather than the best, 3.6 cm. The tracker is weaker than the frame-to-frame odometer (1.9–2.4 cm on those frames), and tracking costs 130–170 ms per frame, reported separately from mapping time and not subject to the TT timer. On fr1_room tracking fails (baseline ATE 0.97 m), so it is shown but excluded from the summary; that threshold (50 cm) was set after seeing the results. On the other four TUM test scenes, TT+VGG raises PSNR at the tracked pose by +0.10 to +2.83 dB (+1.56 dB mean), the same sign as with ground-truth poses, while ATE changes by between −1.61 and +1.50 cm (median −0.63 cm; lower in three of four) on a 6–14 cm baseline, which three seeds cannot separate from zero. The controllers therefore improve map fit at the tracked poses without a reliable effect on this tracker's accuracy. The local Replica sequences are too sparsely sampled for any tracker to work (median step 13–19 cm and 7–11° per frame, versus 0.3–1.1 cm and 0.3–0.7° on TUM; the frame-to-frame odometer gives ATE 1.4–4.7 m), so closed-loop results are TUM only.

Scene	TT+VGG	TT + count cap	Count cap
replica_office0	+1.31	+0.83	−0.01
replica_office1	+0.72	−6.06	−6.35
replica_office3	+1.44	+0.42	−0.32
replica_office4	+1.41	+0.33	−0.25
replica_room0	+0.99	+0.32	−0.23
replica_room2	+1.35	+0.69	−0.03
tum_fr1_room	+0.82	−3.70	−3.69
tum_fr1_rpy	+1.01	−0.48	−0.99
tum_fr1_xyz	+1.12	+0.27	−0.08
tum_fr2_xyz	+3.83	+3.78	+1.68
tum_fr3_sitting_static	+0.06	+0.81	+0.72
mean	+1.28	−0.25	−0.87

Table S22. **Value-gated growth versus a hard map-size budget of the same size.** PSNR change (dB) against the unthrottled baseline, two seeds per scene. The cap on the number of Gaussians equals TT+VGG's final map size on that scene and seed, so the cap is informed by the answer; "TT + count cap" also uses the time-targeted budget.

Value-gated growth versus a map-size budget (Tab. S22). Taming 3DGS and Mini-Splatting control reconstruction through a Gaussian-count budget. We test that control inside our pipeline: a hard cap on the number of Gaussians equal to TT+VGG's final map size on the same scene and seed (so the cap uses information the controller must discover online), with and without the time-targeted budget, two seeds on the 11 test scenes. With the same time-targeted budget the cap averages −0.25 dB against the baseline, against +1.28 dB for TT+VGG, which is better on 10 of 11 scenes (median difference +0.85 dB). The cap fails because it stops growth when the map is full, wherever the new Gaussians would have been needed, whereas VGG decides per frame whether recent spawns were worth keeping; with the time-targeted budget it ends 6.1 dB below the baseline on office1 and 3.7 dB below on fr1_room. These are budget-matched controls, not the published methods, whose scoring and pruning rules we do not reimplement.

L. Controller details

Scene	n	Base	Cap	TT	TT+cap	TT+VGG
office0	4	30.61 (86)	−0.72 (79)	+1.37 (99)	+0.79 (98)	+1.32 (99)
office1	4	33.87 (85)	−1.09 (80)	+0.84 (99)	−0.43 (99)	+0.77 (99)
office3	3	24.40 (88)	−0.46 (77)	+1.41 (100)	+0.86 (96)	+1.42 (100)
office4	3	25.44 (85)	−0.30 (76)	+1.42 (99)	+0.90 (95)	+1.42 (99)
room0	3	22.62 (87)	−0.38 (78)	+1.01 (99)	+0.75 (99)	+0.99 (99)
room2	3	25.55 (87)	−1.67 (77)	+1.40 (99)	−0.55 (99)	+1.33 (99)
fr1_room	3	16.71 (99)	−0.17 (98)	+0.83 (101)	+1.01 (99)	+0.82 (102)
fr1_rpy	3	21.10 (95)	+0.09 (93)	+1.07 (96)	+1.16 (84)	+1.02 (97)
fr1_xyz	4	21.52 (101)	+1.59 (93)	+1.11 (94)	+2.90 (73)	+1.14 (95)
fr2_xyz	4	24.06 (90)	+3.45 (81)	+2.94 (84)	+3.85 (60)	+3.88 (72)
fr3_sitting_static	3	26.40 (69)	+0.11 (55)	+0.06 (56)	+0.09 (55)	+0.05 (57)

Table S23. **Per-scene results, 11 held-out scenes.** Base: unthrottled PSNR (ms); other columns: change in PSNR (dB) with the mean frame time in ms in parentheses.

Scene	n	Base	Cap	TT	TT+cap	TT+VGG
fr1_desk	3	23.09 (92)	−0.18 (90)	+1.02 (82)	+0.47 (65)	+1.04 (82)
fr1_room	3	16.81 (97)	−0.11 (94)	+0.56 (93)	+0.42 (84)	+0.58 (94)
fr1_xyz	3	22.88 (98)	+0.99 (90)	+0.74 (96)	+1.18 (69)	+0.87 (96)
fr2_xyz	3	24.63 (90)	+1.58 (77)	+1.19 (87)	+1.80 (62)	+1.63 (85)
fr3_sitting_static	3	28.35 (63)	+0.50 (50)	+0.34 (55)	+0.47 (51)	+0.41 (56)

Table S24. **Per-scene results, five fresh frame windows.** Base: unthrottled PSNR (ms); other columns: change in PSNR (dB) with the mean frame time in ms in parentheses.

Scene	**Baseline** $\times 1/\times 2/\times 4$	$T = 70$	$T = 85$	$T = 100$	$T = 120$	$T = 140$
fr2_xyz	24.1@88 / 26.5@96 / 27.8@96	27.30@77	27.81@75	27.94@72	27.95@73	27.94@72
fr1_xyz	21.5@100 / 22.8@108 / 24.4@119	21.76@92	21.86@92	22.67@95	23.93@102	24.47@110
office0	30.6@86 / 32.3@105 / 33.4@135	30.64@83	30.79@86	31.93@99	33.00@118	33.43@131
office1	33.8@86 / 34.9@107 / 35.4@138	33.87@84	33.96@86	34.64@99	35.24@119	35.33@133

Table S25. **TT+VGG at five time targets versus fixed budgets.** Mean PSNR (dB) @ mean end-to-end frame time (ms), four seeds. The baseline column gives free growth at modeled budgets $\times 1$, $\times 2$, $\times 4$.